

Prerequisite setup

For using EdgeX and industrial temperature sensor we need the Ubuntu operating system (OS) for Raspberry Pi 4. The latest Ubuntu OS can be downloaded from the link below and the SD card can be prepared by following the steps given therein.

<https://ubuntu.com/tutorials/how-to-install-ubuntu-on-your-raspberry-pi#1-overview>

Ubuntu installation

After preparing the SD card with Ubuntu OS make following changes before booting Raspberry Pi using the micro SD card in which Ubuntu has been installed:

1. To enable USB ports for keyboard, add following line in /boot/config.txt file:

```
max_usb_current = 1
```

2. To enable HDMI ports for display, add following lines in /boot/config.txt file:

```
hdmi_force_hotplug = 1
```

```
hdmi_drive = 2
```

3. To enable SSH, touch the file "ssh" in /boot directory.

After the above changes you can boot the Raspberry Pi by connecting it to 5V DC power adaptor and your TV or monitor using HDMI and providing the internet connectivity using LAN, Wi-Fi, or USB tethering through a mobile phone. Then login with the default credentials (ubuntu/ubuntu) and update the password and get the IP address assigned using command "sudo ip addr". Once you get internet access, you can SSH to Raspberry Pi from another device for remote access.

Next, you need to install the EdgeX on the OS.

There are multiple ways to install EdgeX on Ubuntu. Detailed steps for the installation can be found at <https://wiki.edgexfoundry.org/display/FA/Running+on+a+RaspberryPi>

Snap way of installing EdgeX micro services is easy, but it is a

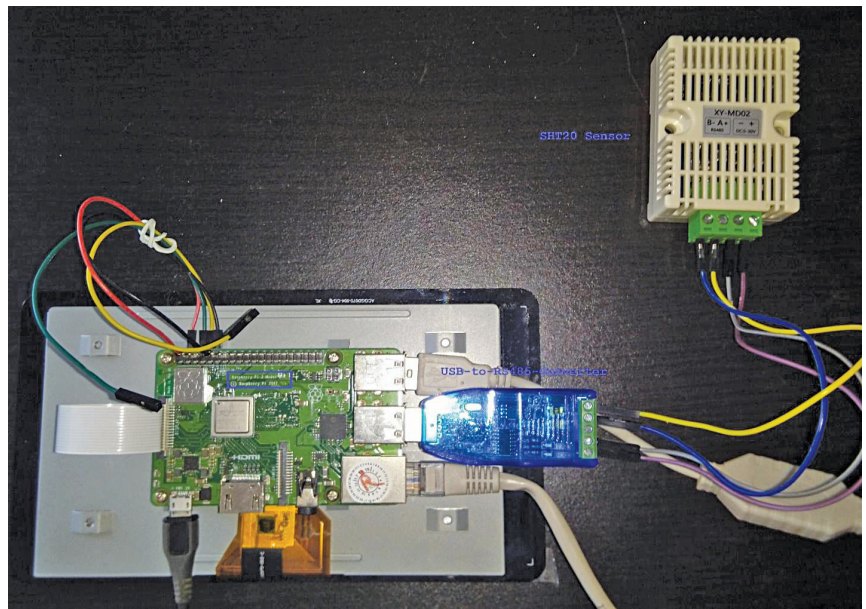


Fig. 3: SHT20 sensor connected to Raspberry Pi using USB-to-RS485 converter

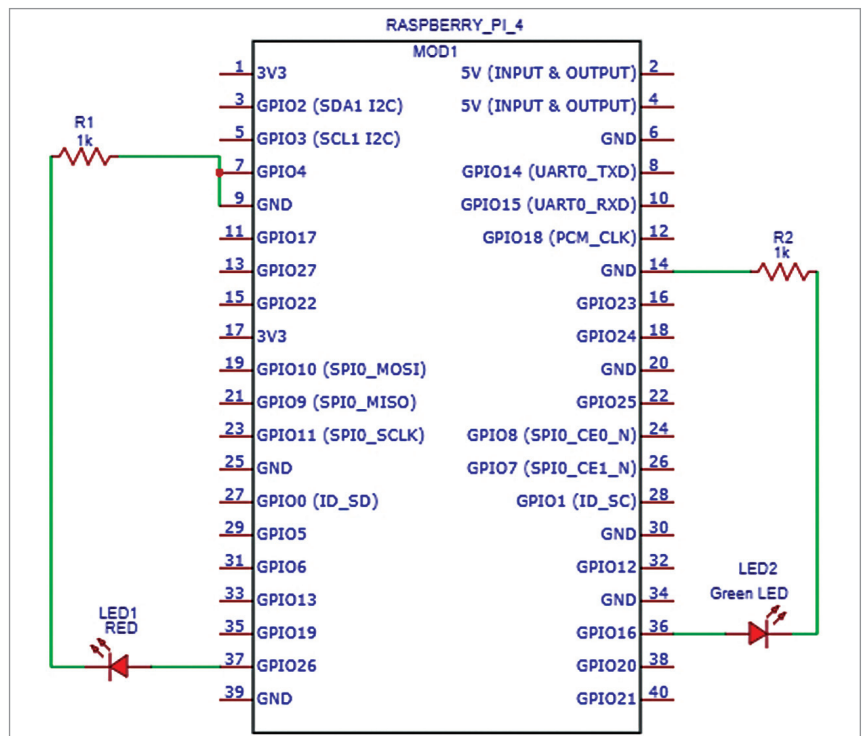


Fig. 4: GPIO pins' connection to LEDs on breadboard

read-only file. So, if there is an issue in debugging any of the microservices, say EdgeX device service, it won't help as you cannot modify the configuration to debug or modify the code. To overcome the read-only problem of Snap and

to debug the code, you can pull the code from GitHub, then build and run.

Sensor device validation

Sensor used for this case study is industrial grade temperature and